PROGRAMME : Diploma Programme in CE / ME / PO / EE / IF / CM / EL / AE / DD / ID / MK
COURSE : Advance Java Programming **COURSECODE** : 233117

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme					Credits	Assessment Scheme											Total Marks
Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL and TSL				Based on self Learning		
											Practical						
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
03	01	02	02	08	04	03	Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
							30	70	100	40	25	10	25#	10	25	10	175

IKS Content: Total Learning Hours for Term: 0 Hrs
# Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

In the current era of networking, online transaction processing and managing the dataflow over network becomes an important issue. This course is essential for providing knowledge and hands on experience over the issues of managing data on Web, developing powerful GUI based friendly user interface, server side programming and developing applications for communication over network using object oriented fundamentals. Advanced Java enhances the Java programming. After learning this course, the student will be able to develop standalone/network/web based software projects required in curriculum as well as industry

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The Aim of this course is to help the students to attain these industry-identified outcomes through various teaching learning experiences:

1. To develop small desktop and web applications
2. Use Advance Java concepts for performing various Server side operations

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Develop applets, Frames using awt and swing components.
2. Develop network based applications.
3. Perform CURD operation on database.
4. Develop server side programs.
5. Develop small desktop or web applications.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I Introduction to Abstract Window Toolkit(AWT) & Swings	1a. Describe use of Frame,Panel and Applets 1b. Use different AWT controls and Layout Managers 1c. Differentiate between AWT and Swing	1.1 Component, container, window, frame, panel, Creating windowed programs & applets. 1.2 AWT controls & layout managers	12

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	1d. Use various Swing components	1.3 Understanding the use of AWT controls: labels, buttons, checkbox, checkbox group, scroll bars, text field, text area 1.4 Understanding the use of layout managers: flowLayout, borderLayout, gridLayout, cardLayout, gridbagLayout, menubars, menus, dialog boxes, file dialog. 1.5 Introduction to swing, Swing features, MVC Architecture, Combo Boxes, Progress Bar	
Unit-II Event Handling	2a. Explain Event delegation model 2b. Explain Event Listener Classes and Interfaces.	2.1 The delegation Event Model, Event sources, Event listeners, Event classes. The Action Event class, The Component Event class, the Container Event class, the Focus Event class, the Item Event class, the Key Event class, the Mouse Event class, the Text Event class, the Window Event class, Adapter classes, Inner classes 2.1 Event listener interfaces, The ActionListener Interface, The ComponentListener Interface, The ContainerListener Interface, the FocusListener Interface, The ItemListener Interface, The KeyListener Interface, The MouseListener Interface, The MouseMotion Interface, The TextListener Interface, the WindowsListener Interface, the WindowFocusListener Interface	10
Unit-III **Networking & Security**	3a. Compare TCP and UDP 3b. Describe Internet Addressing 3c. Explain TCP/IP sockets 3d. Explain URL Connection class and its methods 3e. Write secure coding guidelines for Java	3.1 Basics of Networking: Socket, IP, TCP, UDP, Proxy Server, Internet Addressing 3.2 The InetAddress Class Factory methods, Instance methods 3.3 TCP/IP Sockets, Socket, Server Socket, methods 3.4 URL, URL Connection, http, URL Connection methods, creating & using TCP/IP client & server 3.5 Security with Java: Theoretical introduction to java security, secure coding guidelines for java programming language.	10

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-IV **Interacting with Database**	4a. Explain the role of JDBC and ODBC 4b. Draw JDBC two tier & three tier models, 4c. Establish Connection with Database using JDBC and Retrive information from the database. 4d. Write CRUD operations using JDBC.	4.1 JDBC, ODBC, & Other APIS, JDBC two tier & three tier models, 4.2 Connecting to Database-Driver Interface, Driver Manager Class, Connection Interface, Statement Interface 4.3 The java.sql.package. Establishing connection & retrieving information 4.4 Resultset Interface, CRUD operations using JDBC	12
Unit-V **Servlets, JSP & Spring**	5a. Write uses of Servlet. 5b. Describe Session and Cookies. 5c. Describe use of JSP. 5d. Describe Spring Framework.	5.1 Basics of Web applications, use of tomcat server, Type of Servlet, Servlet life cycle, using servlets, handling request and response. 5.2 Basic concepts of sessions, cookies & session tracking 5.3 JSP introduction, life cycle of JSP, scriptlet tag, expression tag, declaration tag, Implicit Objects, Directives, Action Elements, JSP Expression language, JSTL, JSP custom tags. 5.4 Introduction to Spring Framework- Concept,Features and Advantages of Spring	16
		TOTAL	60

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Sr.No	Unit	UnitTitle	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	**Introduction to Abstract Window Toolkit(AWT) & Swings**	CO1	12	2	4	4	**10**
2	II	**Event Handling**	CO2	10	4	4	8	**16**
3	III	**Networking & Security**	CO3	10	4	4	8	**16**
4	IV	**Interacting with Database**	CO4	12	2	4	8	**14**
5	V	**Servlets, JSP & Spring**	CO5	16	2	4	8	**14**
Grand Total				**60**	**14**	**20**	**36**	**70**

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills **(Outcomes in cognitive, psychomotor and affective domain)** so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	**Unit No.**	**LLO**	**Practical Exercises** (Outcomes in Psychomotor Domain)	**Hours**
01	I	1.a Design a form using AWT components	*Write a program using AWT to design a form using the components textfield, label, checkbox, button, list.	02
02	II	2.a Handle vents related to various AWT components	*Write a program to design a calculator using Java components and handle various events related to each component and apply proper layout to it.	04
03	II	3.a Create a menu bar with various menu items and sub menu	*Write a program to create a menu bar with various menu items and sub menu items. Also create a checkable menu item. On clicking a menu Item display a suitable Dialog box.	02
04	III	4.a Use Combobox to display various items	*Write a program using swing to display a JcomboBox in an applet with the items – cricket, football, hockey, tennis	04
05	III	5.a Implement Jtree and Jtable	*Write a program to create. -Jtree -Jtable	02
06	III	6.a Use Adapter class and its methods	*Write a program making use of Adapter class.	02
07	III	7.a Use InetAddress class methods	*Write a program to retrieve hostname--using methods in Inet Address class.	02
08	IV	8.a Implement client server based chatting program using TCP/IP based communication	*Write a program that demonstrates TCP/IP based communication between client and server.	02
09	IV	9.a Connect to database using JDBC API	*Write an Application program /Applet to make connectivity with database using JDBC API	02
10	IV	10.a Retrive data from database using JDBC	Write an Application program/Applet to send queries through JDBC bridge & handle result.	02

Sr. No.	Unit No.	LLO	Practical Exercises (Outcomes in Psychomotor Domain)	Hours
11	IV	11.a Demonstrate generic servlet class	*Write a servlet for demonstrating the generic servlet class.	02
12	IV	12.a Design a web form to process servlet 12.b Implement cookies 12.c Implement session handling	*Create a web form which processes servlet and demonstrates use of cookies and sessions	02
13	V	13.a Design a user login form using JSP	*Develop a simple JSP program for user login form.	02
14	V	14.a Implement a JSP program to display grade of student	Develop a simple JSP program to display the grade of a student by accepting the marks of five courses.	02
				30

Minimum 12 or more practical's need to be performed out of which the practical's marked as '*' are compulsory.

7.0 Tutorial list

1. Create a List to display multiple items (e.g., list of countries).
2. Create Choice dropdown for selection of cities.
3. Add horizontal and vertical scrollbars. Display real-time values as the user moves the scrollbars.
4. Implement a Java program for Focus Gained and Lost Events with FocusListener
5. Create a Java program for UDP Communication using DatagramSocket and DatagramPacket
6. Write Best Practices for Secure and Efficient JDBC Code
7. Write a program for Navigating ResultSet: First, Last, Previous, Absolute, Relative
8. Explain Cookie versus Session–(Differences,When to Use Cookies,When to Use Sessions, Security Considerations)
9. Elaborate JSP Architecture and Working
10. Spring vs Traditional Java EE

8.0 SELF LEARNING: (Assignment/Activities for Specific Learning/Skill Development/Online Courses/Microprojects)

Other

1. Complete any course of Java Programming on Infosys Springboard/Spoken Tutorial/NPTEL
2. Visit ELearning websites.

Micro project

1. Develop mini-ATM machine system. It should accept account_no, account_holder_name, account_balance and perform operations such as withdrawal, Deposit and balance check.
2. Develop Quiz Management System. Quiz should accept student credentials and contain 10 MCQ type questions. Determine the final result. Save the result in table along with student credentials.
3. Develop Employee Management System. Insert employee details such as employee_name, emp_id, emp_salary etc.into database and retrieve data from table.
4. Any other micro project as suggested by course teacher.

9.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course

1. Massive open online course (MOOCs) may be used to teach various topics/sub topics.
2. Guide micro-projects in group as well as individually.

10.0 ASSESSMENT METHODOLOGY

Formative Assessment-PR
Lab Performance,Viva/voce

Summative Assessment-PR
Lab Performance,Viva/voce

11.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
01	Complete reference	Herbert Schildt	Tata McGraw Hill
02	Unleashed java2 platform	Jamie Jaworkski	Techmedia
03	The Complete IDIOT's guide to JAVA2	Michale Morrison	Prentice Hall of India
04	Java Servlets	Karl Moss	Tata McGraw Hill
05	JSP a beginner's guide	Gray Bolinger and Bharti Natarajan	Tata McGraw Hill
06	Craig Walls	"Spring in Action", Manning"	Dreamtech Press, 4th edition

B) Software/Learning Websites:

1) http://www.oracle.com/technetwork/java/seccodeguide-139067.html
2) http://www.tutorialspoint.com/awt/
3) http://www.javatpoint.com
4) http://docs.oracle.com/javaee/5/tutorial/doc/bnafd.html
5) https://www.tutorialspoint.com/swing/index.htm

C) Major Equipment/ Instrument with Broad Specifications:

1) Computer system - (Any computer system with basic configuration)
2) Software-JDK 1.8 or above, Any IDE for Java Programming such as Eclipse ,JCreator or any other product, Databases like MySql, Oracle, MS-Access, Apache Tomcat Web Server 7 or higher version, Spring Tool Suite, Maven or Gradle

12.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)			
	1	2	3	4	5	6	7	1	2	3	4
CO1	H	H	H					L	L		M
CO2		H	H	M							M
CO3	M	H	H	M			M				M
CO4	M	H	H	M		L					M
CO5		H		H	L		M	H	M	H	M

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

13.0 SUGGESTED QUESTION PAPER PROFILE:

UNIT No	CO	Marks Per Unit	1.35 Times Marks	Question Number Wise Marks						Actual Distribution of Marks
				01	02	03	04	05	06	
I	CO1	10	14	4	4			4	4	16
II	CO2	16	21	4	4	4	4	4		20
III	CO3	16	21	4		4	4	4	4	20
IV	CO4	14	19		4	4	4	4	4	20
V	CO5	14	19	2	4	4	4		4	18
	Total	70	94	14	16	16	16	16	16	94

Table For V units:

Unit No.	I							II							III							IV							V							Total
CO	1							2							3							4							5							
Marks per Unit	10							16							16							14							14							70
1.35 Times marks	14							21							21							19							19							94
Bits	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	Total
CO	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3	4	4	4	4	4	4	4	5	5	5	5	5	5	5	
Q1	2								2	2								2								2								2	2	14
Q2	4								4								4								4											16
Q3	4								4								4								4											16
Q4	4								4								4								4											16
Q5															4								4								4	4				16
Q6								4								4															4	4				16

Sub Total	14	20	22	18	20	94
TOTAL						